

MMALS External Continual-Learning Bridge on SplitCIFAR10: Robust RC2O-v2.2c Evidence over Frozen Visual Feature Hosts

Guillaume Harbonnier
MMALS Project

Robust addendum release: June 11, 2026

Abstract

This report updates the MMALS arXiv/GitHub companion package with the RC2O-v2.2c robust SPLITCIFAR10 addendum. The experiment is designed as an external continual-learning bridge: images are first mapped by a supervised A1 visual backbone into a frozen 256-dimensional feature representation, then the downstream MMALS/RC2O continual-learning stack is evaluated on five disjoint class-incremental tasks. The robust run spans five seeds, eight epochs per task, and validation-selected baseline hardening against replay, EWC, LwF, PNN, and MoE-style controls. Selected RC2O reaches 0.91395 ± 0.00639 final average accuracy, compared with 0.73860 ± 0.01559 for the best replay baseline. The paired gain against replay is $+0.17535$ with 95% confidence interval $[0.16093, 0.18977]$. The selected policy also reports min-task accuracy 0.8535, average forgetting 0.02425, min-class recall 0.8410, and no zero-recall classes. The result supports a robust external continual-learning bridge over frozen supervised visual features. It does not claim end-to-end raw-pixel continual representation learning, CIFAR10 state of the art, or broad external-dataset generality.

1 Purpose and claim boundary

The purpose of this update is to freeze the RC2O-v2.2c SPLITCIFAR10 robust bridge result and integrate it into the paper and GitHub package accompanying the main MMALS and RC2O articles. The claim boundary is deliberately narrow. The experiment tests the continual-learning, routing, memory, readout, and policy-selection layers of MMALS once a usable visual representation has been established. It does not test whether the visual backbone itself can be learned sequentially from raw pixels without forgetting.

Table 1: Supported and unsupported claims.

Status	Claim boundary
Supported	Robust external SplitCIFAR10 CL bridge over frozen A1 visual features.
Supported	Selected RC2O beats validation-selected replay over five seeds.
Supported	Readout collapse is not observed for the selected policy: no zero-recall classes.
Supported as audit	Specialization diagnostics are materialized and aligned to the selected candidate.
Not claimed	End-to-end raw-pixel continual representation learning.
Not claimed	SOTA CIFAR10 performance or broad external-dataset generality.
Not claimed	Positive backward transfer as a demonstrated mechanism.

2 Scientific demarche

The experimental path began with weak raw-flat and early small-CNN SplitCIFAR runs. This created a serious confound: a poor continual-learning score could have come from a weak visual representation rather than from the MMALS/RC2O mechanism. The robust package therefore follows a disciplined sequence: diagnose the representation bottleneck, harden the baselines, validate the A1 visual backbone, run the class-incremental CL protocol, and close the audit loop with selected-policy specialization diagnostics.

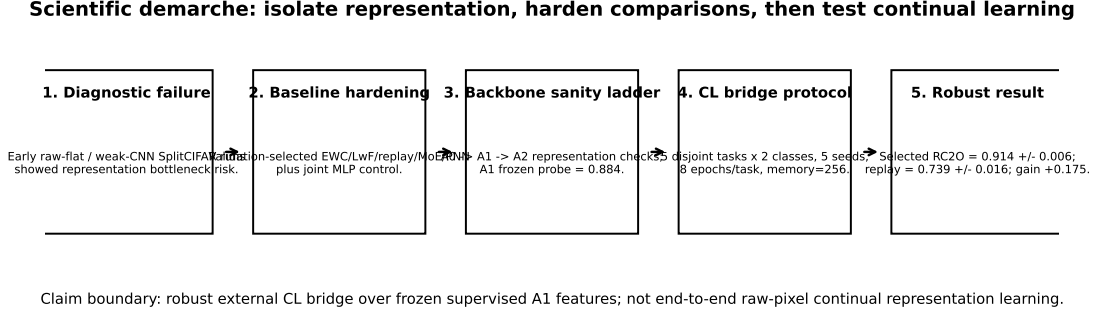


Figure 1: Scientific demarche used to move from representation debugging to robust external continual-learning evidence.

3 Architecture: perceptual host and functional hosts

The current experiment separates the perceptual host from the downstream functional MMALS hosts. Let x denote a raw CIFAR10 image. A1 is a perceptual host

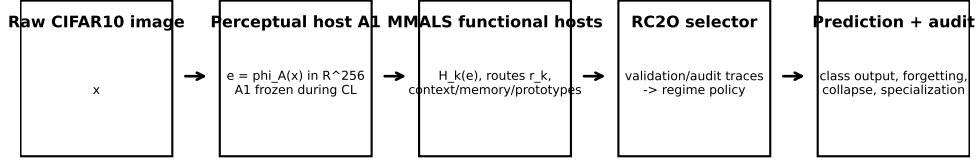
$$e = \phi_A(x; \theta_A) \in \mathbb{R}^{256}, \quad (1)$$

trained and sanity-checked before the continual-learning sequence. During the CL experiment, θ_A is frozen. The downstream MMALS hosts then operate on e :

$$z_k = H_k(e), \quad r_k = q_k(e, m, c), \quad z = \sum_k r_k z_k, \quad \hat{y} = h(z). \quad (2)$$

The RC2O controller is a policy selector over validation and audit-memory traces. It does not simply report the best final-test row; it selects a deployable family under regime and safety constraints.

Perceptual host vs functional MMALS hosts



Current v2.2c: A1 is trained and sanity-checked before CL, then frozen. MMALS/RC2O performs continual learning over A1 features.

Figure 2: Current v2.2c architecture: A1 is a frozen perceptual host; MMALS functional hosts, memory and policy selection perform continual learning over A1 features.

4 Protocol

The robust package uses five disjoint binary tasks over CIFAR10 classes:

$$(0, 1) \rightarrow (2, 3) \rightarrow (4, 5) \rightarrow (6, 7) \rightarrow (8, 9). \quad (3)$$

After each new task, all seen tasks are evaluated. Thus the evaluation chain is the standard continual-learning pattern: train on a new task, infer on old and new tasks, update memory and prototypes, then continue to the next task. The robust run uses seeds $\{0, 1, 2, 3, 4\}$, eight epochs per task, and memory size 256. Baseline hyperparameters are selected on validation memory, not by arbitrary defaults.

5 Backbone sanity ladder

The representation question is resolved by the A1 backbone sanity ladder. The A1 supervised backbone and A1 frozen linear probe both reach approximately 0.884 in robust mode. This makes the downstream CL comparison meaningful: if MMALS failed after this point, the most plausible source would be memory, readout, routing or selector behavior, not a visibly broken visual feature extractor.

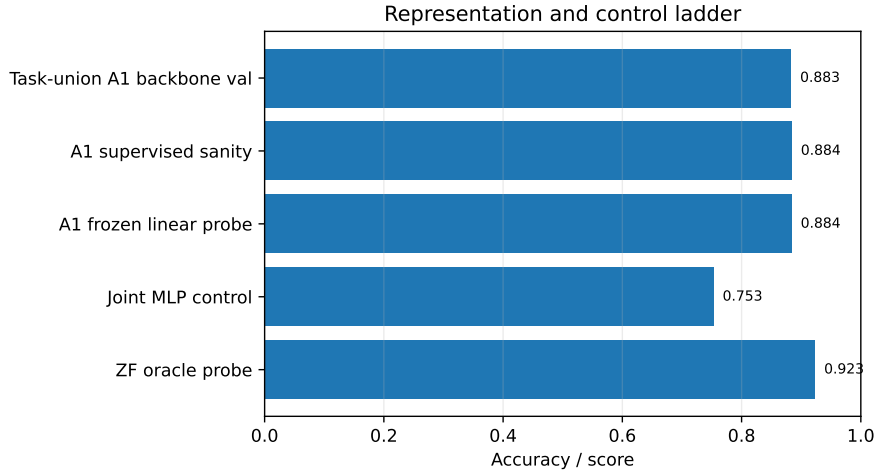


Figure 3: Representation and control ladder in the robust package. The row named joint upper bound in the raw notebook is interpreted here as a joint MLP control, not as a theoretical upper bound.

Table 2: Representation ladder and controls.

Stage	Metric	Value	Secondary	Diagnostic
Task-union A1 backbone val	feature_backbone_val_acc	0.8831	0.8987	yes
A1 supervised sanity	val_acc_best	0.8844	0.8797	yes
A1 frozen linear probe	val_acc_best	0.8841	0.8803	yes
Joint MLP control	final_avg_acc_mean	0.7531	—	no
ZF oracle probe	probe_proto_zf_oracle	0.9234	—	yes

6 Robust results

Table 3 reports the robust 5-seed leaderboard. Selected RC2O is close to the ZF oracle probe and clearly above the best replay baseline. The safe readout candidate is essentially tied with the selected policy, which is useful because it indicates that the selected policy is not a brittle outlier.

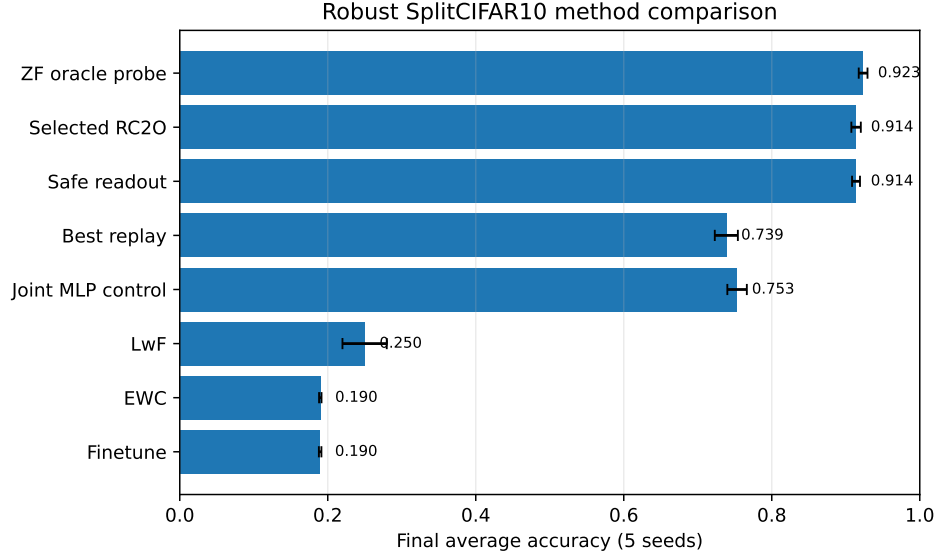


Figure 4: Robust method comparison over five seeds. Error bars denote standard deviation where available.

Table 3: Robust method leaderboard.

Method	Final avg. acc.	Min-task	Forgetting	Min-class	Size MB
Selected RC2O	0.9140 ± 0.0064	0.8535	0.0242	0.8410	2.011
Safe readout	0.9139 ± 0.0054	0.8530	0.0239	0.8430	2.011
RC1b guarded context gap	0.9097 ± 0.0081	0.8470	0.0298	0.8285	2.011
ZF oracle probe	0.9234 ± 0.0059	0.8678	—	0.8270	0.002
ZF proto probe	0.9206 ± 0.0086	0.8642	—	0.8140	0.002
Best replay	0.7386 ± 0.0156	0.6020	0.1921	0.5665	0.316
Joint MLP control	0.7531 ± 0.0132	0.6245	0.0000	0.5795	0.316
Finetune	0.1897 ± 0.0017	0.0000	0.9306	0.0000	0.316
EWC	0.1899 ± 0.0016	0.0000	0.9346	0.0000	0.316
LwF	0.2498 ± 0.0299	0.0380	0.0039	0.0005	0.316

6.1 Per-seed stability

The paired selected-RC2O gain over replay is positive in every seed. The average gain is $+0.17535$ with 95% confidence interval $[0.16093, 0.18977]$. The oracle gap remains small, averaging 0.0095.

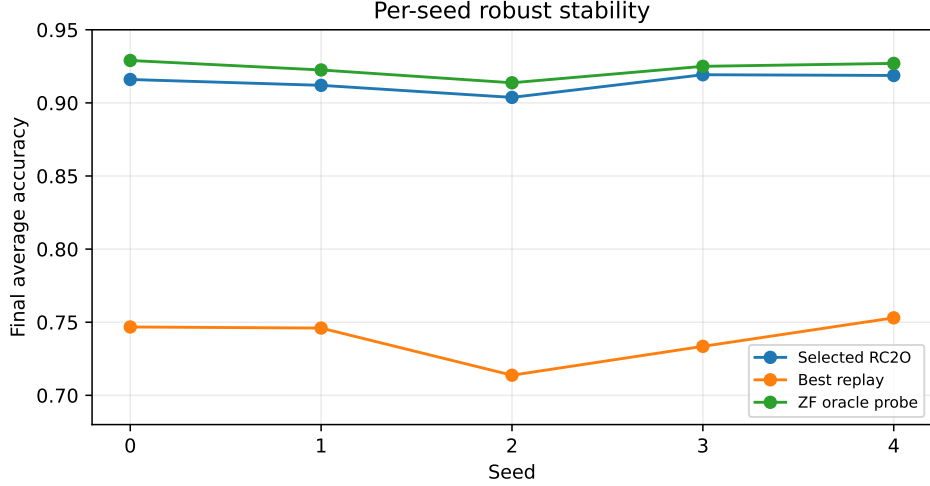


Figure 5: Per-seed accuracy of selected RC2O, best replay and the oracle probe.

Table 4: Per-seed robust comparison.

Seed	Selected	Safe	Replay	Joint ctrl.	Oracle	Δ vs replay	Oracle gap
0	0.9160	0.9130	0.7468	0.7605	0.9290	0.1693	0.0130
1	0.9120	0.9120	0.7460	0.7465	0.9225	0.1660	0.0105
2	0.9038	0.9062	0.7137	0.7627	0.9137	0.1900	0.0100
3	0.9193	0.9193	0.7335	0.7330	0.9250	0.1857	0.0057
4	0.9187	0.9187	0.7530	0.7630	0.9270	0.1657	0.0083

7 Baseline hardening

A central reviewer risk in earlier runs was that replay, EWC, LwF, PNN or MoE controls could be dismissed as arbitrary defaults. The v2.2c robust package addresses this by exporting a validation-memory baseline hardening trace. For each seed, baseline candidates are selected by validation score before the final comparison. This makes the baseline comparison stronger, though it is still not a claim of state-of-the-art baseline tuning.

Table 5: Validation-selected baseline hardening summary.

Baseline family	Selected seeds	Val. score	Val. final avg	Val. min-task	Val. forgetting
balanced_replay	5	0.7238	0.6973	0.5577	0.1171
experience_replay	5	0.7238	0.6973	0.5577	0.1171
lwf	5	0.2289	0.2285	0.0190	0.0063
sparse_moe	5	0.0066	0.1759	0.0000	0.6769
pnn	5	0.0062	0.1746	0.0000	0.6737
noisy_topk_moe	5	0.0059	0.1751	0.0000	0.6765
ewc	5	0.0051	0.1753	0.0000	0.6806
hard_top1_moe	5	0.0050	0.1729	0.0000	0.6715
dense_moe	5	0.0043	0.1737	0.0000	0.6777
finetune	5	0.0040	0.1733	0.0000	0.6771
joint_upper_bound	5	—	—	—	—

8 Readout collapse, min-class behavior and forgetting

The selected policy reports min-task accuracy 0.8535, average forgetting 0.02425, min-class recall 0.8410, and no zero-recall classes. These values matter because early SplitCIFAR runs were often dominated by class collapse or task collapse. The robust result is therefore not merely a high average-accuracy row; it is also a non-collapse result.

9 Selected-policy specialization audit

The v2.2c patch fixed a previous audit gap: route/context, host ablation, host probe and prototype separation are now aligned to the selected policy candidate when available. The robust run materializes all four components across seeds.

Table 6: Specialization audit materialization and selected-policy alignment.

Component	OK seeds	Rows	Total rows	Aligned
host_ablation	5	5	150	yes
host_probe	5	5	125	yes
prototype_separation	5	5	50	yes
route_context_and_confusion	5	5	50	yes

10 SplitCIFAR10 external-bridge gate

The raw notebook exports a SplitCIFAR10-specific external-bridge gate. This gate is not a MNIST-family robust gate; it checks whether the bridge experiment is interpretable: A1 passes representation sanity, the joint MLP control is no longer collapsed, selected RC2O is competitive with hardened replay, min-task and min-class metrics are non-collapsed, and specialization components are present.

Table 7: SplitCIFAR10 external-bridge gate.

Check	Value	Floor	Ceiling	Status
a1_supervised_backbone_pass	0.8844	0.6500	–	PASS
a1_frozen_linear_probe_pass	0.8841	0.5500	–	PASS
joint_upper_bound_representation_floor	0.7531	0.5500	–	PASS
selected_rc2o_competitive_with_replay	0.9140	0.7286	–	PASS
selected_min_task_nonzero	0.8535	0.0500	–	PASS
selected_forgetting_not_explosive	0.0242	–	0.2000	PASS
selected_min_class_nonzero	0.8410	0.0500	–	PASS
selected_zero_recall_classes_gate	–	–	0.0000	PASS
specialization_components_materialized	4.0000	4.0000	–	PASS
specialization_aligned_to_selected_policy	1.0000	1.0000	–	PASS

11 Discussion

This result upgrades SPLITCIFAR10 from a representation-debugging branch to a robust external CL bridge. The strongest scientific contribution is not that MMALS is now a universal

continual-learning solution. The contribution is narrower and stronger: once the representation bottleneck is controlled and baselines are validation-hardened, the MMALS/RC2O stack robustly preserves high accuracy and low forgetting over the SplitCIFAR10 sequence.

Two caveats are central. First, the visual backbone is frozen during continual learning. This isolates the downstream memory and routing layers, but it does not validate end-to-end raw-pixel continual representation learning. Second, the row originally named joint upper bound should be renamed joint MLP control. Because selected RC2O exceeds it, it cannot be interpreted as a mathematical upper bound. It is a useful control with the same readout scale, not a ceiling.

12 Limitations and next steps

The immediate next step is not to expand the claim, but to repeat the same discipline on harder data. The recommended sequence is: freeze this SPLITCIFAR10 robust bridge; update the main MMALS and RC2O companion papers; then move to SplitCIFAR100, CORe50, TinyImageNet or DomainNetMini with the same baseline-hardening and representation-ladder discipline. A stronger future version would promote the A1 perceptual host into a slow-plasticity host, constrained by feature-stability, prototype-stability and context-consistency losses.

13 Reproducibility package

The GitHub package contains the LaTeX source, compiled PDF, figures, tables, robust CSV/JSON summaries, selected raw CSV artifacts, the v2.2c notebook, and the robust analysis report. The included README documents the build command and claim boundary.